

Summary of Laws – Economics Midterm

Module 1: Introduction to Economics

- Law of Demand: Price $\uparrow \rightarrow$ Demand $\downarrow$ (inverse relationship). Diagram: Downward sloping demand curve. Exceptions: Giffen goods, Veblen goods, necessities, addiction, etc.
- Law of Supply: Price $\uparrow \rightarrow$ Supply $\uparrow$ (direct relationship). Diagram: Upward sloping supply curve.

Module 2: Welfare Analysis

- Law of Diminishing Marginal Utility (DMU): More consumption $\rightarrow$ less extra satisfaction. Explains downward slope of demand curve.
- Law of Substitution (part of substitution effect): When a good becomes cheaper, it replaces more expensive goods in consumption.

Module 3: Production and Cost Function

- Law of Variable Proportions (Short Run): As variable input increases with fixed inputs, output rises first at increasing rate, then diminishing, then negative returns.
- Law of Returns to Scale (Long Run): When all inputs change proportionally, output may increase more, equal, or less than proportionately (IRTS, CRS, DRTS).
- Law of Equi-Marginal Principle: Resources allocated such that marginal benefit/product per unit cost is equal across all uses. Basis of consumer equilibrium and cost minimization.

Quick Recap Table

Module	Law	Key Idea	Curve/Diagram
M1	Law of Demand	Price $\uparrow \rightarrow$ Demand $\downarrow$	Downward demand curve
M1	Law of Supply	Price $\uparrow \rightarrow$ Supply $\uparrow$	Upward supply curve
M2	Law of DMU	More consumption $\rightarrow$ less extra satisfaction	MU curve downward
M2	Law of Substitution	Cheaper goods replace expensive ones	Shift in demand
M3	Law of Variable Proportions	3 phases of output with variable input	TP, AP, MP curves
M3	Law of Returns to Scale	Output change with proportional input change	Isoquants
M3	Law of Equi-Marginal	Equalize marginal returns across uses	Tangency condition



Summary of Laws – All Modules

Module 1: Introduction to Economics

1. Law of Demand

Definition: Other things constant, quantity demanded $\uparrow$ when price $\downarrow$, and vice versa.

Diagram: Downward sloping demand curve.

Reasons: Income effect, substitution effect, diminishing marginal utility.

Exceptions: Giffen goods, Veblen goods, future expectations, necessities, ignorance, addiction, brand loyalty.

2. Law of Supply

Definition: Other things constant, quantity supplied $\uparrow$ when price $\uparrow$, and vice versa.

Diagram: Upward sloping supply curve.

Reason: Higher prices attract more production.

Module 2: Welfare Analysis

3. Law of Diminishing Marginal Utility (DMU)

Definition: As a consumer consumes more units of a good, the additional satisfaction (marginal utility) from each successive unit decreases.

Diagram: Utility curve slopes downwards after a point.

Importance: Explains downward slope of demand curve & consumer choices.

4. Law of Substitution (Implied in SE)

Definition: When the price of a good falls, it becomes relatively cheaper, so consumers substitute it for other

goods.

Related Concepts: Income Effect + Substitution Effect
→ explain consumer behavior.

Module 3: Production and Cost Function

5. Law of Variable Proportions (Short Run Production Function)

Definition: When one factor (e.g., labor) is increased while others fixed, output first increases at an increasing rate, then diminishing rate, and finally decreases.

Phases:

Increasing returns (efficiency, specialization).

Diminishing returns (overcrowding of variable factor).

Negative returns (too much variable input reduces output).

Diagram: Total Product, Average Product, and Marginal Product curves.

6. Law of Returns to Scale (Long Run Production Function)

Definition: When all inputs are varied in the same proportion, output may increase more than proportionately (increasing returns), exactly proportionately (constant returns), or less than proportionately (decreasing returns).

Diagram: Isoquants showing scale expansion.

7. Law of Equi-Marginal Principle (Applied in Production & Consumption)

Definition: A consumer (or producer) allocates resources such that the marginal utility (or marginal product) per unit of cost is equal across all uses.

Application: Optimal consumption, cost minimization.

✓ Quick Recap Table:

Module	Law	Key Idea	Curve/Diagram
M1	Law of Demand	Price ↑ → Demand ↓	Downward sloping demand curve
M1	Law of Supply	Price ↑ → Supply ↑	Upward sloping supply curve
M2	Law of DMU	More consumption → less extra satisfaction	MU curve ↓
M2	Substitution Law	Cheaper goods replace expensive ones	SE curve shift
M3	Law of Variable Proportions	Short-run input increase → 3 phases of output	TP, AP, MP curves
M3	Law of Returns to Scale	Long-run proportional input change → IR, CR, DR	Isoquants
M3	Law of Equi-Marginal	Equalize marginal returns across uses	Tangency condition

